

BARONS AGCM, AB, Canada

WMO: 710070

Lat: 50.026N

Lon: 113.224W

Elev: 967

StdP: 90.23

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-30.3	-26.7	-33.7	0.2	-30.3	-29.7	0.3	-26.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	30.7	15.7	28.7	15.2	26.9	14.8	18.3	25.5	17.3	24.9	16.2	24.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	12.6	21.7	14.5	11.6	20.1	13.3	10.7	18.7	56.0	25.2	52.3	24.8	49.1	24.5	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A		-35.8	34.6	3.2	2.1	-38.1	36.1	-40.0	37.4	-41.8	38.6	-44.1	40.1
			DB	-35.9	21.0	3.2	1.4	-38.2	22.0	-40.0	22.8	-41.8	23.6	-44.1	24.6
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-6.9	-7.6	-1.5	4.3	10.4	14.3	17.5	17.0	12.2	5.3	-1.6	-8.3
	DBStd	11.17	8.66	8.76	7.81	5.09	4.21	3.10	2.56	3.17	4.59	5.41	7.83	8.73
	HDD10.0	2712	523	493	357	178	47	3	0	1	29	160	352	569
	HDD18.3	5041	782	726	614	422	246	124	47	63	188	403	599	827
	CDD10.0	763	0	0	1	7	61	134	231	217	95	15	3	0
	CDD18.3	50	0	0	0	0	1	5	19	21	4	0	0	0
	CDH23.3	1521	0	0	0	10	94	152	513	544	202	4	0	0
	CDH26.7	453	0	0	0	1	17	35	153	187	59	1	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	408	10	13	20	27	60	100	50	41	38	18	16	12
	PrecMax	658	22	36	47	52	109	289	120	94	138	43	44	29
	PrecMin	179	1	0	4	9	8	44	8	2	5	4	1	0
	PrecStd	107	6	10	13	12	32	59	35	23	32	10	11	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	12.8	16.3	24.7	28.7	31.0	32.9	33.6	31.1	23.3	18.2	10.6
		MCWB	5.4	6.0	7.2	10.8	12.9	16.6	17.0	16.1	14.1	11.1	8.6	4.2
	2%	DB	8.2	9.0	13.9	19.9	26.1	27.1	30.3	30.9	28.4	20.1	13.9	7.6
		MCWB	3.4	3.8	5.9	8.7	12.4	14.5	16.7	15.5	13.5	10.2	7.0	3.2
	5%	DB	6.3	6.5	11.7	16.9	23.2	24.8	28.1	28.7	25.7	17.2	11.1	5.5
		MCWB	2.4	2.0	4.9	7.2	11.5	14.3	16.3	15.3	13.0	8.8	5.6	1.8
	10%	DB	4.7	4.4	9.1	13.9	20.4	22.8	26.2	26.5	22.5	14.7	8.5	3.8
		MCWB	1.4	0.9	3.5	5.8	10.3	13.1	16.2	14.9	11.9	7.5	3.8	0.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	6.5	8.3	11.0	13.9	18.9	20.8	18.5	15.9	12.6	9.4	5.6
		MCDB	10.5	11.9	14.6	23.1	24.6	25.8	26.8	26.9	25.9	21.1	16.8	9.5
	2%	WB	3.8	4.1	6.7	9.2	12.9	16.7	19.0	17.4	14.4	10.8	7.6	3.5
		MCDB	7.8	8.9	12.8	18.4	24.1	24.5	26.2	26.3	25.1	18.8	13.5	6.9
	5%	WB	2.6	2.3	5.2	7.7	11.9	15.3	17.9	16.5	13.4	9.2	5.8	2.1
		MCDB	5.9	5.9	11.1	15.8	21.9	22.6	24.6	25.6	23.8	16.4	10.6	5.1
	10%	WB	1.5	0.9	3.8	6.2	10.9	14.1	16.9	15.7	12.4	7.8	4.1	0.7
		MCDB	4.6	4.1	8.8	13.1	19.0	20.9	23.9	24.6	21.9	14.2	8.4	3.8
Mean Daily Temperature Range	5% DB	MDBR	11.6	12.5	12.4	13.8	14.0	13.8	15.7	16.4	15.7	14.1	12.2	11.6
		MCDBR	12.2	14.6	15.6	19.9	20.0	18.3	20.3	21.4	22.2	17.9	14.1	12.4
		MCWBR	8.4	9.7	8.9	9.8	8.1	8.0	8.9	8.4	9.1	9.2	8.4	8.7
	5% WB	MCDBR	11.4	14.2	15.0	18.5	17.8	16.1	16.6	18.0	19.9	17.0	14.0	12.3
		MCWBR	8.1	10.0	8.9	9.4	7.5	7.8	8.5	8.0	8.5	9.1	8.6	9.1
Clear-Sky Solar Irradiance	taub	0.252	0.260	0.283	0.320	0.332	0.329	0.333	0.348	0.311	0.281	0.258	0.233	
	taud	2.442	2.482	2.480	2.387	2.404	2.460	2.462	2.401	2.481	2.534	2.497	2.450	
	Ebn at Noon	821	907	937	927	925	928	918	886	891	867	809	799	
	Edh at Noon	59	73	88	108	111	106	104	106	87	69	55	50	
All-Sky Solar Radiation	RadAvg	1.23	2.22	3.51	4.84	5.77	6.37	6.97	5.56	3.94	2.44	1.35	0.95	
	RadStd	0.10	0.19	0.28	0.33	0.45	0.44	0.35	0.35	0.34	0.26	0.12	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page